

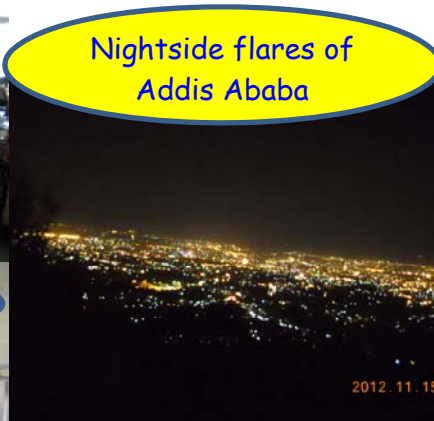
Welcome reception



Lunch time socialization



Nightside flares of
Addis Ababa



Field trip by the bus



Axum and Lalibela



Entertainment at the
dinner reception



Meeting concluded by
minster of S & T.



Deputy Prime Minister
opened the meeting



Poster Session



Greeting from Lucy



The panelist



The Coffee ceremony



Chapman Conference on longitudinal and hemispheric dependence of space weather

Despite the fact that much progress has been made in the study of ionospheric density structure and dynamics in the last decade, there are many gaps remain in the global understanding of the fundamental electrodynamics that governs the formation of equatorial ionospheric density irregularities. This is in part due to the uneven distribution of ground-based instruments, largely due to the lack of ground-based instrumentation in Africa. Until recently, ionospheric density structure in the African sector has been only estimated by model interpolation over vast geographic areas. However, satellite data have shown that the equatorial ionospheric density structures in the African continent, respond to space weather effects differently than do other parts of the Earth. However, these unique responses have not been confirmed, validated, or studied in detail due to the lack of suitable ground-based instrumentation in the region. Thus, the cause of these unique density irregularities in the continent, including their temporal and spatial evolutions, remains a mystery for the scientific community. Therefore, to gain a better understanding, almost eight years ago the scientific communities, under the umbrella of United Nations–sponsored International Space Weather Initiative (ISWI) the then International Heliophysical Year (IHY), showed significant interest to close the largest land-based gap in ground-based instruments on the continent which was devoid of ground-based instruments (see left panel in [Figure 1](#)). Because, eight years ago when we look at a map of the world that shows the location of ground-based space physics instrumentations (radars, magnetometers, ionosondes, GPS dual frequency receivers, and others), one can quickly recognize Africa’s lack of space physics research infrastructure. Thus during the last couple of years, quite significant number of small-to-medium scale instruments have been either deployed or in the process of deployment in Africa as shown in [Figure 1](#) (right panel). In addition to new scientific discoveries and advancing space science research in Africa by establishing scientific collaborations between scientists in developed and developing nations, the deployment of those instruments spontaneously increase the number of African space scientists as well as boost the importance of space science awareness in many African countries.

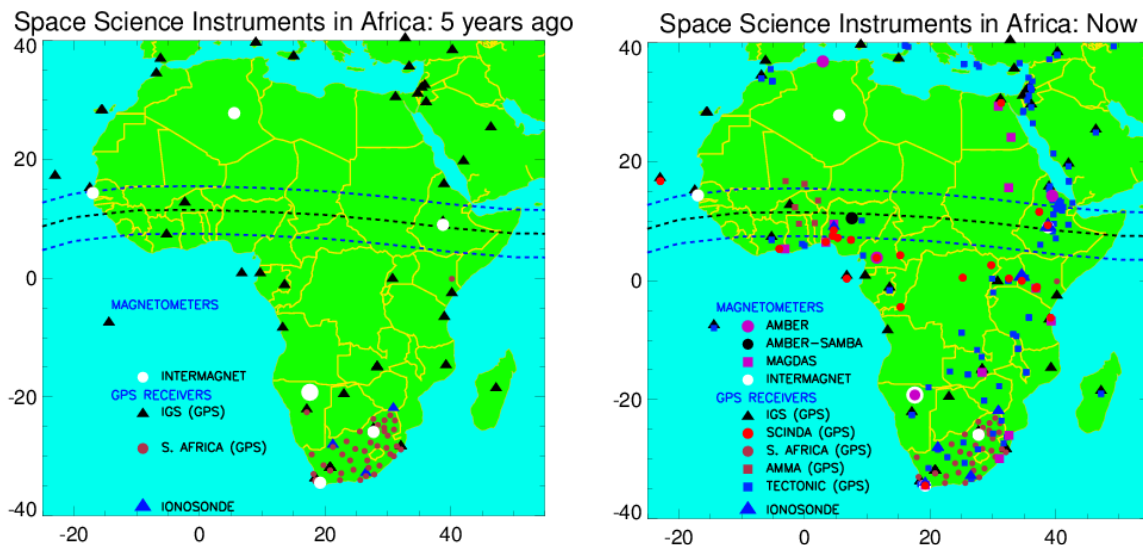


Figure 1. (left panel) Ground-based instrument coverage in African five years ago, and (right panel) shows the current ground-based instrumentation in Africa. Yellow big circle is the proposed site.

The scientific communities also recognize that in order to develop space science research infrastructure within Africa, space science educational infrastructure also needs to be developed to support the long-term operation and use of the science instrumentation. One way to address this concern is to increase and facilitate a continuous strong interaction between scientists from developed countries and African young professors and postgraduate students. In order to create such interaction between the two parties, the space science communities organize idealistic and scientific meetings and held in Africa. In response to these needs, several international workshops/conferences and summer schools were already conducted in different part of Africa. The very successful IHY-Africa 2007 (held in Addis Ababa, Ethiopia in November 2007), IHY-Africa 2009 workshop (held in Livingstone, Zambia in June 2009), IHY summer school (held in Abuja, Nigeria in 2009) and ISWI summer school (held in Bahir Dar, Ethiopia in 2010) are among those that had been held in Africa. More recently, in recognition to the unprecedented achievement of the space science development in Africa, the American Geophysical Union (AGU) decided and held one of its prestigious Space Science meetings, Chapman conference, in Africa for the first time.

Attendees and Logistics

The AGU Chapman conference on longitudinal and hemispheric dependence of space weather, which was the first space weather Chapman conference in the African continent, has been held successfully in Addis Ababa, Ethiopia during 12-16 November 2012. The conference, which was organized primarily by Colorado University and Boston College (in collaboration with Addis Ababa University, Bahir Dar University, and Ethiopian Space Science Society), become possible through three committees; the conveners, the International Programme Committee, and the Local Organizing Committee (LOC). A complete list of members of these committees appears in Appendix A. The conference has been well attended by excess of 110 participants (see *Figure 2*), represented from more than 27 nations. Of these delegates about 45% represented from more than 12 different African countries, about 35% were from USA and Canada, and the remaining 20% were from other countries that include Europe, Asia, South America and Australia. Because of the generosity of the sponsors, about 74 African delegates and 15 delegates from outside African were sponsored either partially or in full. Travel arrangements for the selected sponsored African scientists were made using a travel agency in the USA and Italy coordinated by Boston College, Colorado University, and ICTP. The travel arrangements for the sponsored US scientists and other delegates outside African and USA were also made in the US through Boston College and Colorado University. The conference participants, including those who received full or partial sponsorship, were accommodated in two hotels (Intercontinental Addis and Jupiter International Hotels) that are close to Economic Commission for Africa (ECA) hall, the conference venue. The LOC in collaboration with the conveners coordinated all the accommodation arrangements with the hotels. *Figure 3* shows the geographic distribution of countries around the world represented at Chapman conference in Ethiopia. The list of countries represented at the workshop is given in Appendix C, and a complete list of delegates can be found in Appendix D.



Figure 2: Group photo of the participants take outside the meeting hall. List of sponsors' logs are shown at the top and bottom side of the picture.

The conference was held at United Nation for Economic Commission of Africa (UN-ECA) meeting hall (see Figure 4). The opening ceremony was attended by higher government officials that include the Honorable Ethiopian deputy Prime Minister and Minister of Education, Honorable Demeke Mekonnen and Honorable Ethiopian Minister of Science and Technology, Honorable Dessie Dalike. At the opening ceremony several presentations were made by different agency representatives, including Dr Sunanda Basu (AGU's representative), Dr R. Behnke (NSF's representative), Dr. M. Guhathakurta (NASA's representative), Dr. N. Gopalswamy (SOCTEP's representative), and others.

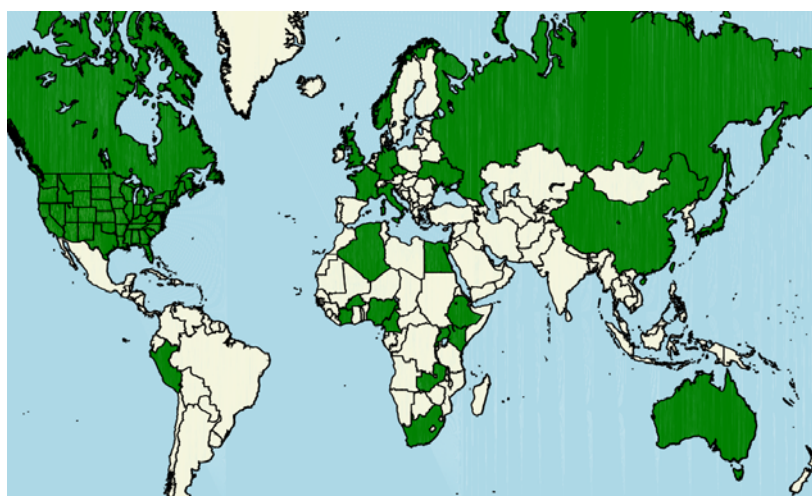


Figure 3: Geographic distribution of countries (shaded in green) that were represented at Chapman Conference in Addis Ababa, Ethiopia



Figure 4: The venue where Chapman conference was held, the United Nations for Economic Commission for Africa (UN-ECA).

Scientific Program

The conference has been organized for two prime objectives: (1) to assemble an international group of heliophysics scientists to plan and discuss current and needed observations at mid and low latitudes in the African longitude sector, a region that has never been explored in detail using ground-based instruments, and (2) to enhance the space science education and research interest in the Africa continent. The first half day (Monday morning) were dedicated to the opening ceremony followed by presentation by three major African space agencies (South African, Nigerian and Ethiopian space agencies), presentation about the legacy and achievements of the International Heliophysical Year (IHY) and International Space Weather Initiative (ISWI), and presentation about the scientific exploration using GNSS systems in Africa. In addition to the Infrastructure development and panel discussion sessions, the scientific programme of the conference included six main science themes:

1. Hemispherical Dependence of Magnetospheric Energy Injection and the Thermosphere-Ionosphere Response
2. Longitude and Hemispheric Dependence of Storm-Enhanced Densities (SED)
3. Response of the Thermosphere and Ionosphere to X-Ray and EUV Time-History During Flares
4. Quiet-Time Longitude Spatial Structure in Total Electron Content and Electrodynamics
5. Temporal Response to Lower-Atmosphere Disturbances, and
6. Ionospheric Irregularities and Scintillations.

Very compelling and high standard oral and poster papers, for each scientific theme, were presented at the conference. For all five science sessions as well as for infrastructure developments session there were about 65 oral presentations, with the main focus on observation and modeling efforts of understanding the unique space weather events in the African region. It is worth to mention that about 34% of the total oral presentations made at the conference were presentations by African scientists. This indicates how high standard science contribution from African scientists significantly increased compared to the previous meetings. In addition, the conference had one poster sessions as well as one panel discussion on Infrastructure and Capacity Building. The poster sessions, which included about 40 poster presentations, were divided according to the main science sessions mentioned above. The panel discussion, which includes seven panelist (F. Anad (Algeria), L. McKinnell (South Africa), C. Amory-Mazaudier (France), J. Adeniyi (Nigeria), S. Tessema (Ethiopia), S. Radicella (Italy), and K. Groves (USA))

and chaired by T. Fuller-Rowell and Su. Basu, was designed to find a way forward to address the most important question not only for African scientist but also for the international scientific communities in general, i.e., what type of observations and activities are needed in the region to address the outstanding science questions, such as why African sector ionospheric irregularities are so unique and active all year round compared to other longitudinal sectors?

Finally, on Friday morning, following the panel discussion, the conference concluded by formally establishing the African Geophysical Society (AGS) that was officially inaugurated at the closing ceremony of Chapman Conference. This will be remembered as another mile stone contribution of AGU's Chapman conference in Addis Ababa, which served as a vehicle that brought the African scientist into one roof and provide them the opportunity to have a final discussion and officially announced for the birth date of AGS.

Establishment of AGS: the idea of establishing AGS popped up at the informal discussion among African scientists who attended the first SCINDA meeting that was held in Cape Verde back in July 2006. Since then a small self initiated groups organized themselves and made a significant steps toward the formation of AGS, and the first report about the framework of AGS were presented by Dr. B. Rabiou at IHY-Africa 2009 meeting in Livingstone. Quite a lot of brainstorm discussions among African scientists have been floating around mainly focused on suggestions how to proceed forward. At the inaugural ceremony the African participants also elected committee members who will lead the society for the next two years; and they are (1) Prof Babatunde Rabiou from Nigeria - President, (2) Dr Ayman Mahrous from Egypt - Vice President, (3) Dr Solomon Belay from Ethiopia - Secretary General, (4) Dr Cesar Mbane from Cameroon - Assistant Secretary General, (5) Prof Olivier Obrou from Cote d'Ivoire - Treasurer.

Conclusion Remarks: The Chapman conference was successful in all aspect. In additions being a venue of scientific brainstorm discussion, the meeting served as a vehicle and brought African Space Scientists together and providing a forum in which they could meet and make contacts with international colleagues in the field. A number of collaborations have been established as a result of such similar conferences and workshops. More importantly, quite a number of students were exposed to the Space Science field and had opportunity to have a one-on-one discussion with successful scientists in the field (see Figure 5 & 6 below). It is very imperative that meetings like Chapman conference can form a very valuable part of scientific research not only for Africans but also for scientists from different part of the world, and should be highly encouraged and supported.

The Chapman conference would not have been possible without the support of the sponsors, like NSF, NASA, ICTP, ONR, NOAA, SCOSTEP, and would not have been successful without the support of the Ethiopian government and local sponsors (Addis Ababa University, Bahir Dar University, Ethiopian Space Science Society, Ministry of Education, and Ministry of Science and Technology of Ethiopia). The complete list of sponsors is shown in Appendix E. The conveners are also thankful for the support and hard work of local organizing team.

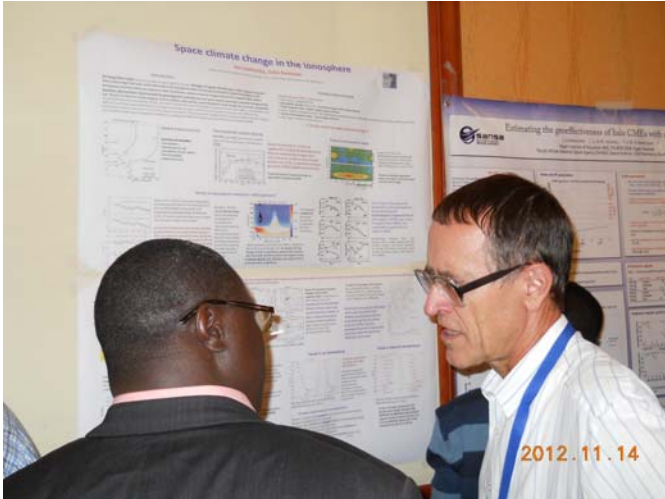


Figure 5: African professor discussing his results with scientist from Europe (left panel) and an African student having private discussion with scientist from USA about his research work (right panel).

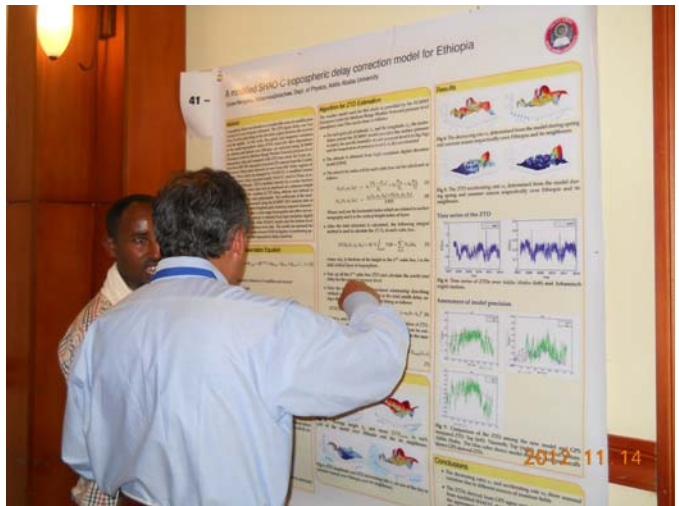


Figure 6: Left panel shows an African student socializing with scientist from USA, and right panel depicts an African student discussing his poster presentation with scientist from USA about his research work (right panel).

Appendix A

List of Conveners and Organizing Committees

Conveners:

1. **Tim Fuller-Rowell**, University of Colorado and NOAA/SWPC, USA
2. **Endawoke Yizengaw**, Boston College, USA
3. **Patricia Doherty**, Boston College, USA
4. **Gizaw Mengistu**, Addis Ababa University, Ethiopia

International Program Committee:

1. **Jacob Adeniyi**, University of Ilorin, Nigeria
2. **Scott Bailey**, Virginia Tech, USA
3. **Paul Baki**, Kenya Polytechnic University College, Kenya
4. **Sunanda Basu**, Boston College and the American Geophysical Union, USA
5. **Keith Groves**, Boston College, USA
6. **Rod Heelis**, University of Texas at Dallas, USA
7. **Mike Kelley**, Cornell University, USA
8. **Lee-Anne McKinnell**, South African National Space Agency, South Africa
9. **Sandro Radicella**, International Center for Theoretical Physics, Italy
10. **Kiyohumi Yumoto**, Kyushu University, Japan
11. **Eftyhia Zesta**, Air Force Research Laboratory, USA

Local Organizing Committee (LOC) – Ethiopia:

1. **Baylie Damtie**, Bahir Dar University (LOC Chairman)
2. **Gizaw Mengistu**, Addis Ababa University (LOC Vice Chairman)
3. **Solomon Tessema**, Ethiopian Space Science Society (LOC Secretary)
4. **Alemu Abebe**, Ethiopian Ministry of Science and Technology (Member)
5. **Solomon Shiferaw**, Ethiopian Ministry of Education (Member)

Appendix B

PROGRAM

Time	Title	Presenter	Chairs
Sunday (11 November 2012)			
5:00PM-7:00PM	Registration and Welcome Reception—Intercontinental Addis Hotel		
Monday (12 November 2012)			
Opening Ceremony			
8:00AM-8:45am	Late Registration (Please try to register Sunday evening)		
Begin 9:00AM	Opening Remarks by LOC Chair	B. Damtie	Mengistu
9:05AM-9:10AM	Opening remarks by Nigerian Minister of Science and Technology		
9:10AM-9:15AM	Opening remarks by South African Minister of Science and Technology		
9:15AM-9:30AM	Implementing AGU's vision, mission and strategic plan: efforts to date and future opportunities	C. McEntee (Su. Basu)	
9:30AM-9:45AM	ISWI/SCOSTEP contribution to the development of space science in Africa	N. Gopalswamy	
9:45AM-10:00AM	AMISR in Africa	R. Behnke	
10:00AM-10:15AM	Role of NASA-ILWS to the African space science development	M. Guhathakurta	
10:15AM-10:25AM	Remarks from the ICTP	S. Radicella	
10:25AM-10:40AM	Conference Opening Speech by Ethiopian Deputy Prime Minister and Minister of Education	H. E. Mr. Demeke Mekonnen	

10:40AM-11:10AM	Coffee Break		
African Space Agency Activities			
11:10AM-11:30AM	An overview of the South African National Space Agency (SANSA) <i>(Invited)</i>	L. McKinnell	Mengistu
11:30AM-11:50PM	Nigerian Space Agency <i>(Invited)</i>	Minster Sci. & Technology	
11:50PM-12:10PM	The experience of space science and astronomy education and research in Ethiopia <i>(Invited)</i>	S. Tessema	
12:10PM-12:30PM	Some achievements of the International Heliophysical Year (IHY) and International Space Weather Initiative (ISWI) <i>(Invited)</i>	C. Amory-Mazaudier	
12:30PM-12:50PM	Scientific exploration using GNSS systems in Africa <i>(Invited)</i>	P. Doherty	
12:50PM-2:00PM	Lunch Break		
Hemispherical Dependence of Magnetospheric Energy Injection and the Thermosphere-Ionosphere Response			
2:00PM-2:30PM	Hemispherical dependence of magnetospheric energy injection <i>(Invited)</i>	E. Zesta (M. Moldwin)	Fuller-Rowell/ Yizengaw
2:30PM-2:50PM	Asymmetry in hemispheric characteristics in Sq behavior over Africa within latitudes 29°N and 35°S	A. B. Rabiou	
2:50PM-3:10PM	Differences in ionospheric response to magnetic disturbances at the northern and southern hemispheres	J. Laštovička	
3:10PM-3:30PM	Ionospheric response, characterized by variations in GPS derived TEC monitored at Lagos, Nigeria, to the intense storm of 9 th March 2012	L. Amaeshi	

3:30PM-3:50PM	Physical implications for discrepancy between the summer and winter PC indices observed in the course of magnetospheric substorms	O. Troshichev	
3:50PM-4:20PM	Coffee Break		
4:20PM-4:50PM	Global-scale ultraviolet imaging of the hemispheric and longitudinal dependence of the thermosphere and ionosphere (<i>Invited</i>)	L. Paxton	
4:50PM-5:20PM	Tomographic imaging of ionospheric electron density over Eastern Africa using multi-platform instrumentations and model ionospheric total electron content (<i>Invited</i>)	G. Mengistu Tsidu	Rabiu/ Doherty
5:20PM-5:40PM	Daytime altitude and longitude variations of the equatorial, topside magnetic field-aligned ion transport at solar minimum	A. Burrell	
5:40PM-6:00PM	A real-time model of the equatorial ionospheric zonal electric field	M. Nair	
Evening	Open		

Tuesday (13 November 2012)

Longitude and Hemispheric Dependence of Storm-Enhanced Densities (SED)

8:30AM-9:00AM	Longitude variations in storm enhanced densities (<i>Invited</i>)	R. Heelis	
9:00AM-9:20AM	Explaining the very intense geomagnetic storm of November 20-21, 2003, and the pre-storm phenomenon	V. Chukwuma	Damtie/ Bailey
9:20AM-9:40AM	Observations of SED during the storm of August 3-4, 2010	Su. Basu	

9:40AM-10:00AM	The response of African equatorial GPS-TEC to intense geomagnetic storms during the ascending phase of solar cycle 24	A. Akala	
10:00AM-10:20AM	A global ionospheric range error correction model for single frequency GNSS users	N. Jakowski	
10:20AM-10:50AM	Coffee Break		
10:50AM-11:20AM	CEDAR system science: Geospace thermal plasma redistribution-longitude and hemispheric dependence of storm-enhanced densities, SAPS, and geomagnetic storms <i>(Invited)</i>	J. Foster	McKinnell/ Heelis
11:20AM-11:40AM	Statistical study of the Equatorial F_2 layer at Ouagadougou during solar cycles 20, 21 and 22, using Legrand and Simon's classification of geomagnetic activity	F. Ouattara	
11:40AM-12:00PM	The ionospheric F_2 -layer and magnetic storm of May 29th, 2010: A comparism between Ilorin, Hermanus and Jicamarca	B. Joshua	
12:00PM-12:20PM	Closing the geophysical data gap- Midlatitude ionospheric characterization using GPS TEC time and position variability derived from GPS measurements in Zambia	P. Sibanda	
12:20PM-12:40PM	Solar Cycle 24 observations of storm enhanced density and the tongue of ionization	A. Coster	
12:40PM-2:00PM	Lunch Break		
Response of Thermosphere and Ionosphere to X-Ray and EUV Time-History During Flares			
2:00PM-2:30PM	Spectrally-resolved X-Ray and extreme ultraviolet irradiance variations during	T. Woods	Moldwin/ Doubmbia

	solar flares <i>(Invited)</i>		
2:30PM-3:00PM	The ionospheric response to solar EUV variability <i>(Invited)</i>	P. Anderson	
3:00PM-3:20PM	Effects of radio burst on GPS measurements over Africa	O. Obrou	
3:20PM-3:40PM	Simultaneous observation of ionospheric irregularities over the African region	C. Ngwira	
3:40PM-4:00PM	Coffee Break		
4:00PM-4:30PM	Tracing solar energy through the upper atmosphere: the response of lower thermospheric nitric oxide to solar flares and the impact of this response on the ionosphere <i>(Invited)</i>	S. Bailey	Obrou/ Valladares
4:30PM-4:50PM	Signatures of Anatolian bump in ITNE-CIDR of Egypt	A. Mahrous	
4:50PM-5:10PM	Effects solar transients on the ionospheric ionization and their positional dependence	S. Tripathi	
5:10PM-5:30PM	Equatorial electrojet ground induced currents: Are there power grid space weather impacts at equatorial latitudes?	M. Moldwin	
6:30PM-9:30PM	Poster Session—Jupiter International Hotel Refreshments served		
Wednesday (14 November 2012)			
Quiet-Time Longitude Spatial Structure in Total Electron Content and Electrodynamics			
8:30AM-9:00AM	Determining the sharp, longitudinal gradients in equatorial $\mathbf{E} \times \mathbf{B}$ drift velocities associated with the boundaries of the 4 - cell, non-migrating structures <i>(Invited)</i>	D. Anderson	Baki/ Caton

9:00AM-9:20AM	Variability of Total Electron Content and the IRI model predictions over equatorial stations	E. Oyeyemi	
9:20AM-9:40AM	On the variabilities and uncertainties in the measurement of absolute (true) TEC over Indian equatorial and low latitude sectors	P. Rao	
9:40AM-10:00AM	Characteristics of the Equatorial F_2 ionospheric layer at low solar activity period <i>(Invited)</i>	J. Adeniyi	
10:00AM-10:20AM	Modeling the equatorial electric field in the African sector using ground magnetic observatory data	P. Alken	
10:20AM-10:50AM	Coffee Break		
10:50AM-11:20AM	Wave forcing from the lower atmosphere <i>(Invited)</i>	E. Talaat	Ouattara/ Amory- Mazaudier
11:20AM-11:50AM	The climatological and day-to-day longitudinal variability of the global ionospheric density distribution <i>(Invited)</i>	E. Yizengaw	
11:50AM-12:10PM	On the longitudinal variation of the equatorial electrojet, its dependence on the geomagnetic main field intensity <i>(Invited)</i>	V. Doumbia	
12:10PM-12:30PM	The effects of $\mathbf{E} \times \mathbf{B}$ drifts on the equatorial ionosphere during extreme solar minimum	J. Klenzing	
12:30PM-1:30PM	Lunch Break		
2:00PM-6:00PM	Field Trip		
Evening	Open		

Thursday (15 November 2012)

Temporal Response to Lower-Atmosphere Disturbances			
8:30AM-9:00AM	Ionospheric response to lower atmospheric disturbances and impact on space weather <i>(Invited)</i>	L. Goncharenko	Mahrous/ Coster
9:00AM-9:30AM	Ionospheric electrodynamics response at low latitudes to lower-atmosphere disturbances <i>(Invited)</i>	J. Chau	
9:30AM-9:50AM	Observations of TIDs over South America	C. Valladares	
9:50-10:10AM	Gravity wave dynamics and mesospheric temperature variability over Africa-A plan for new measurements	P. Loughmiller	
10:10AM-10:30AM	Comparison of thermospheric winds and temperatures measured simultaneously in Peru and Brazil by Fabry-Perot interferometers	J. Meriwether	
10:30-11:00AM	Coffee Break		
11:00AM-11:30AM	Modeling and predicting the space weather response to terrestrial weather <i>(Invited)</i>	T. Fuller-Rowell	Chau/D. Anderson
11:30AM-11:50AM	Forecasting auroral radio absorption from the epsilon parameter	O. Ogunmodimu	
11:50AM-12:10PM	Ionosphere characterization at low latitudes using TEC global maps	S. Radicella	
12:10PM-12:30PM	Validation of IRI-2007 and NeQuick2 and TEC ingestion into NeQuick 2 to model East-African equatorial Ionosphere	M. Nigussie	
12:30PM-12:50PM	Quasi 16-day periodic meridional movement of Equatorial Ionization Anomaly	D. Zhang	
12:50PM-2:00PM	Lunch Break		

Ionospheric Irregularities and Scintillation				
2:00PM-2:30PM	Longitudinal variations of low latitude irregularities and scintillation <i>(Invited)</i>	K. Groves	Tessema/ Foster	
2:30PM-2:50PM	Low latitude ionospheric scintillation and ionospheric irregularity Drifts observations w/ GPS-SCINDA and VHF receiver in Kenya	J. Olwendo		
2:50PM-3:10PM	C/NOFS Observations of Longitudinal Ionospheric Variability	O. de la Beaujardière		
3:10PM-3:30PM	Day-to-day and seasonal TEC variability time-shift near the Equatorial Ionospheric Anomaly region over Africa <i>(Invited)</i>	P. Baki	Mbane/ Jakowski	
3:30PM-3:50PM	Source variability of strong convection influence on the upper atmosphere and ionosphere	R. Friedel		
3:50PM-4:20PM	Coffee Break			
4:20PM-4:50PM	Deep gravity wave dynamics and responses in the thermosphere and ionosphere <i>(Invited)</i>	D. Fritts	Mbane/ Jakowski	
4:50PM-5:20PM	Modeling of the longitudinal dependence of equatorial plasma bubbles <i>(Invited)</i>	J. Huba		
5:20PM-5:40PM	Development of an ionospheric map for Africa	N. Ssessanga		
5:40PM-6:00PM	Impact of Arctic and Antarctic ionospheric scintillations on GNSS	S. Skone	Mbane/ Jakowski	
7:00PM-11:00PM	Conference Dinner Banquet Hiber Ethiopia National Restaurant			
Friday (16 November 2012)				
Infrastructure Development				

8:30AM-8:45AM	Equatorial ionospheric scintillation effects on GNSS and SATCOM: Real-time observations with the Scintillation Network Decision Aid (SCINDA) <i>(Invited)</i>	R. Caton	Adeniyi/ Groves
8:45-9:00AM	MAGDAS and other magnetometer networks in the African continent and some of their scientific benefits <i>(Invited)</i>	G. Maeda	
9:00AM-9:15AM	Overview of European Space Agency (ESA) <i>(Invited)</i>	N. Jakowski	
9:15AM-9:30AM	Overview about AFREF network <i>(Invited)</i>	A. Nonguierma	
9:30AM-9:45AM	AfricaArray—An environmental monitoring network and capacity building initiative for Africa <i>(Invited)</i>	A. Wondem	
9:45AM-10:15AM	Coffee Break		
Panel Discussion			
10:15AM-12:15AM	Infrastructure and Capacity Building: What observations and activities are needed in the region to address the outstanding science questions? 5 minute opening statements followed by open discussion	F. Anad L. McKinnell C. Amory-Mazaudier J. Adeniyi S. Tessema S. Radicella K. Groves	Fuller-Rowell/ Basu
12:15PM-12:30PM	Wrap up closing remarks	Fuller-Rowell/ Basu/Doherty	
12:30PM-12:45PM	Final Closing remarks Ethiopian Minister of Science and Technology	H.E Mr. Dessie Dalike	
12:45PM-2:00PM	Lunch Break		

Open time for side meetings			
Afternoon	Open for side meetings		

Appendix C

List of Countries represented at the meeting

1. Algeria
2. Australia
3. Burkina Faso
4. Cameroon
5. Canada
6. China
7. Czech Republic
8. Egypt
9. Ethiopia
10. France
11. Germany
12. India
13. Italy
14. Ivory Coast
15. Japan
16. Kenya
17. Nigeria
18. Norway
19. Peru
20. Russia
21. Rwanda
22. South Africa
23. Uganda
24. UK
25. Ukraine
26. USA
27. Zambia

Appendix D

List of Participants and their contact information

Last Name	First Name	Affiliation	Country	E-mail
Abdulrahim	Rasheedat	National Space Res. & Dev. Agency	Nigeria	bourlarnley@gmail.com
Abebe	Alemau	Ethiopian Ministry of Science and Technology, Addis Ababa	Ethiopia	alemuabebe2002@yahoo.com
Abraha	Gebregiorgis	Addis Ababa Univ., Addis Ababa	Ethiopia	ggabraha@yahoo.com.hk
Adediji	Adekunle	Federal Univ. of Technology, Akure	Nigeria	kunleadediji2002@yahoo.co.uk
Adejuyigbe	Oluwaloseyi	Univ. of Lagos, Lagos	Nigeria	comiforte@yahoo.com
Adekoya	John	Olabisi Onabanjo Univ., Ago Iwoye	Nigeria	adekoyabolarinwa@yahoo.com
Adeniyi	Jacob	Univ. of Ilorin, Ilorin	Nigeria	segun47@yahoo.com
Akala	Andrew	Univ. of Lagos, Lagos	Nigeria	akalaovie2004@yahoo.com
Alken	Patrick	Univ. of Colorado, Boulder, CO	USA	alken@colorado.edu
Amaeshi	Larry	Univ. of Lagos, Lagos	Nigeria	larryamma@yahoo.co.uk
Amory-Mazaudier	Christine	LPP/Polytechnique/UPMC/CNRS, Saint-Maur-des-Fossés	France	christine.amory@lpp.polytechnique.fr
Anad	Fatma	CRAAG, Algiers	Algérie	f_anad@yahoo.fr
Anderson	David	CIRES, Univ. of Colorado (NOAA/SWPC), Boulder, CO	USA	david.anderson@noaa.gov
Anderson	Phillip	Univ. of Texas at Dallas, TX	USA	phillip.anderson1@utdallas.edu
Arunpold	Sanit	AIT, Bangkok	Thailand	sanit.arunpold@gmail.com
Ayalew	Eliase	Addis Ababa Univ., Addis Ababa	Ethiopia	eliaseayalew@ymail.com
Ayalew	Tibebu G.	Bahir Dar Univ., Bahir Dar	Ethiopia	tib2010get@gmail.com
Ayokunnu	Olalekan D.	The Polytechnic, Ibadan	Nigeria	davilek@yahoo.com
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Appendix E

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